

# **Optimizing Turnaround Time and Unlocking Off-Season Efficiency: A Dual Data-Driven Approach for Service Center Growth**

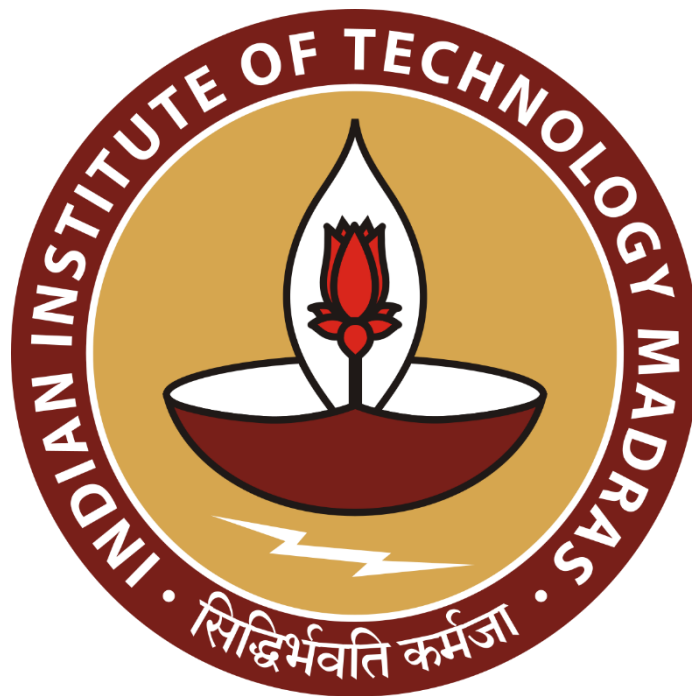
**Midterm Report for the BDM capstone Project**

Submitted by

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# 1 Executive Summary

## **Title: Optimizing Turnaround Time and Unlocking Off-Season Efficiency: A Dual Data-Driven Approach for Service Center Growth**

Prime Infotech, operating as an authorized warranty service center in Calicut, Kerala, plays a critical role in bridging brands and end users by efficiently managing warranty claims for top brands such as Gigabyte and ADATA. Over the past months, the company has encountered two significant challenges: delays in the warranty claim turnaround (TAT) and reduced technician utilization during the summer season and it leading to less revenue generation.

Analysis of operational and claim records reveals that inconsistent TAT across brands often leads to SLA breaches, delayed brand payments, and diminished process reliability. Seasonal dips in service volume further contribute to idle capacity and fluctuating revenue, impacting the center's profitability and its relationships with partner brands. The current report conducts descriptive assessment of TAT and SLA status for each brand, pinpoints trends in breach frequency and monthly service throughput, and evaluates periods of low resource utilization.

By quantifying performance across major brands and highlighting significant seasonal effects, this midterm report provides a data-driven foundation to recommend practical solutions. The findings serve as actionable guidance for sustained business improvement and smarter operational planning going forward.

## 2 Proof of originality of the Data



Figure 1: Front view of the office

**Business Details:**

Prime Infotech

Owner Name: Sajeev Kumar T

Contact: 8129823260

Address: Safa Building, Ram Mohan Rd, near Bairangikulam GLP School, Chindhavalappu, Palayam, Kozhikode, Kerala 673004



**Interaction video with the owner:**

[https://drive.google.com/file/d/1IWAC\\_sJxCXbtP92DujLXcY8YqarF10aw/view?usp=drive\\_link](https://drive.google.com/file/d/1IWAC_sJxCXbtP92DujLXcY8YqarF10aw/view?usp=drive_link)

**Data Link:**

<https://docs.google.com/spreadsheets/d/11bpRpoKzRCvrQMl2YxyypmuQZLPnPyRH/edit?usp=sharing&oid=104741596898347695757&rtpof=true&sd=true>

**More images from the office:** [Prime Infotech images](#)

### 3 Metadata and Descriptive Statistics

My collected dataset contains detailed records of product RMAs (Return Material Authorizations) for Gigabyte and ADATA (the two major brands with date-wise claim records) over 11 months from September 2024 to July 2025. The data was initially received in multiple sheets, which was combined into one then matched and corrected dates, and manually filled in any missing values to ensure completeness and accuracy.

| PRIME INFOTECH CLOSE CALL DATA FOR THE MONTH-DECEMBER- 2024 |                 |                      |                             |      |             |                     |                     |                     |               |
|-------------------------------------------------------------|-----------------|----------------------|-----------------------------|------|-------------|---------------------|---------------------|---------------------|---------------|
| RMA NO                                                      | Cust Name       | Product Name         | RMA SN                      | Duty | Engineer ID | Receive Date        | Repair Date         | Test Date           | Shippin       |
| KC02-24A0125                                                | APPLE COMPUTERS | M/B H510M H 1.3B     | 4719331817428SN214950121684 | N    | IN220004    | 2024-10-28 18:45:39 | 2024-10-29 19:06:32 | 2024-10-30 14:15:11 | 2024-11-06 09 |
| KC02-24B0055                                                | SHYLESH         | M/B H410M H V2 1.3A  | 4719331815387SN223250085754 | C    | IN220004    | 2024-11-11 16:51:22 | 2024-11-13 12:33:28 | 2024-11-14 13:46:49 | 2024-11-18 10 |
| KC02-24B0104                                                | Yunax computers | M/B H410M H V3 1.0A  | 4719331823542SN212950038231 | VS   | INPT2404    | 2024-11-19 13:25:25 | 2024-11-27 13:35:21 | 2024-11-28 14:27:03 | 2024-12-02 13 |
| KC02-24B0107                                                | SYSCON SYSTEM   | M/B H410M S2 V3 1.3A | 4719331823559SN22057A143548 | C    | IN230021    | 2024-11-19 14:28:13 | 2024-11-27 13:55:51 | 2024-11-28 15:42:44 | 2024-12-02 13 |
| KC02-24B0116                                                | macbizz         | M/B H410M S2 V2 1.4A | 4719331815424SN223550074470 | C    | IN220004    | 2024-11-21 19:30:11 | 2024-11-29 19:49:45 | 2024-11-29 20:30:18 | 2024-12-04 10 |
| KC02-24B0125                                                | SVR Computers   | M/B GA-H110M-H 1.1B  | 4719331847210SN224870295835 | VS   | INPT2404    | 2024-11-23 18:21:55 | 2024-11-30 09:40:01 | 2024-11-30 17:04:31 | 2024-12-03 10 |
| KC02-24B0129                                                | INFO SIGHT      | M/B H410M H V3 1.2A  | 4719331823542SN22203A011138 | C    | IN220008    | 2024-11-26 19:19:45 | 2024-11-30 12:48:12 | 2024-11-30 12:48:29 | 2024-12-03 10 |
| KC02-24B0130                                                | INFO SIGHT      | M/B H410M H V3 1.2A  | 4719331823542SN22203A011137 | C    | IN220008    | 2024-11-26 19:21:12 | 2024-11-29 14:25:08 | 2024-11-30 11:55:56 | 2024-12-03 10 |
| KC02-24B0131                                                | INFO SIGHT      | M/B H410M H V3 1.2A  | 4719331823542SN22203A011136 | C    | IN220008    | 2024-11-26 19:22:00 | 2024-11-29 14:26:16 | 2024-11-30 12:37:05 | 2024-12-03 10 |

Figure 2: Sample of original monthly RMA service data

| Send Date | courier     | RMA NO         | Column1            | Column2   | Column3                     | Column4           | SN                          |
|-----------|-------------|----------------|--------------------|-----------|-----------------------------|-------------------|-----------------------------|
| 8/1/2024  | CLT30331604 | KC02G-B2470044 | D9                 | ADATA SSD | 14460218                    | 2M422L281KY1      |                             |
| 8/1/2024  | CLT30331604 | KC02-2470165   | H310M H            | 84733020  | 4719331803445SN231160056821 | H310M H           | 4719331803445SN231160056821 |
| 8/1/2024  | CLT30331604 | KC02-2470167   | H510M H V2         | 84733020  | 4719331854942SN23343A028561 | H510M H V2        | 4719331854942SN23343A028561 |
| 8/1/2024  | CLT30331604 | KC02-2470169   | GP-P550B           | 85044090  | SN23303G007069              | GP-P550B          | SN23303G007069              |
| 8/1/2024  | CLT30331604 | KC02-2480001   | GP-GSTFS31240G NTD |           | SN211508921999              | GP-GSTFS31240GNTD | SN211508921999              |
| 8/1/2024  | CLT30331604 | KC02-2480002   | H510M S2           | 84733020  | 4719331817398SN215260077650 | H510M S2          | 4719331817398SN215260077650 |

Figure 3: Separately maintained monthly Send Date records

The final dataset contains 14 columns and of about 1017 cases, providing a comprehensive overview of warranty service operations.

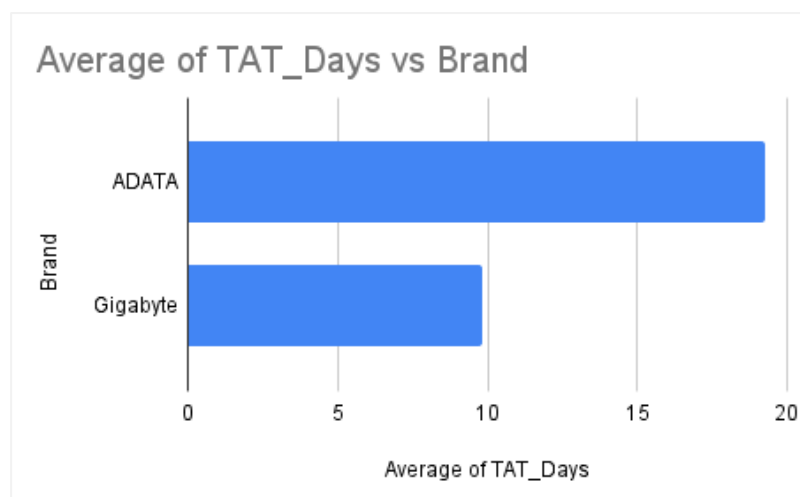
| RMA NO       | Product Name             | RMA SN                      | Brand    | Receive Dat | Repair Dat | Shipping Dat | Close Dat  | Send Dat   | Engineer ID | Month  | TAT_Day | Repair |
|--------------|--------------------------|-----------------------------|----------|-------------|------------|--------------|------------|------------|-------------|--------|---------|--------|
| KC02-2480047 | M/B H510M S2 1.0M        | 4719331817398SN21476004633  | Gigabyte | 2024-08-07  | 2024-08-08 | 2024-08-15   | 2024-09-16 | 2024-08-07 | IN220008    | Sep-24 | 40      |        |
| KC02-2480093 | M/B H410M H V3 1.2A      | 4719331823542SN22126004152  | Gigabyte | 2024-08-14  | 2024-09-03 | 2024-09-05   | 2024-09-05 | 2024-08-14 | IN220008    | Sep-24 | 22      |        |
| KC02-2480094 | M/B B560M GAMING HD 1.2C | 4719331818630SN22185003448  | Gigabyte | 2024-08-14  | 2024-09-05 | 2024-09-06   | 2024-09-06 | 2024-08-14 | IN230012    | Sep-24 | 23      |        |
| KC02-2480116 | M/B H510M H 1.3B         | 4719331817428SN21523A02757  | Gigabyte | 2024-08-19  | 2024-08-31 | 2024-09-03   | 2024-09-03 | 2024-08-20 | IN230012    | Sep-24 | 15      |        |
| KC02-2480120 | M/B B560M GAMING HD 1.2C | 4719331818630SN22185003445  | Gigabyte | 2024-08-20  | 2024-09-04 | 2024-09-06   | 2024-09-06 | 2024-08-20 | INPT2404    | Sep-24 | 17      |        |
| KC02-2480141 | M/B H310M S2 1.1A        | 4719331803469SN21436004130  | Gigabyte | 2024-08-22  | 2024-08-28 | 2024-09-02   | 2024-09-02 | 2024-08-23 | IN230021    | Sep-24 | 11      |        |
| KC02-2480149 | M/B H410M S2 V2 1.6A     | 4719331815424SN22464001172  | Gigabyte | 2024-08-23  | 2024-08-31 | 2024-09-03   | 2024-09-03 | 2024-08-23 | IN220004    | Sep-24 | 11      |        |
| KC02-2480151 | M/B H410M H V3 1.2A      | 4719331823542SN22203A011138 | Gigabyte | 2024-08-23  | 2024-09-07 | 2024-09-11   | 2024-09-11 | 2024-08-23 | INPT2404    | Sep-24 | 10      |        |

Figure 4: Preview of final cleaned and consolidated dataset used for analysis

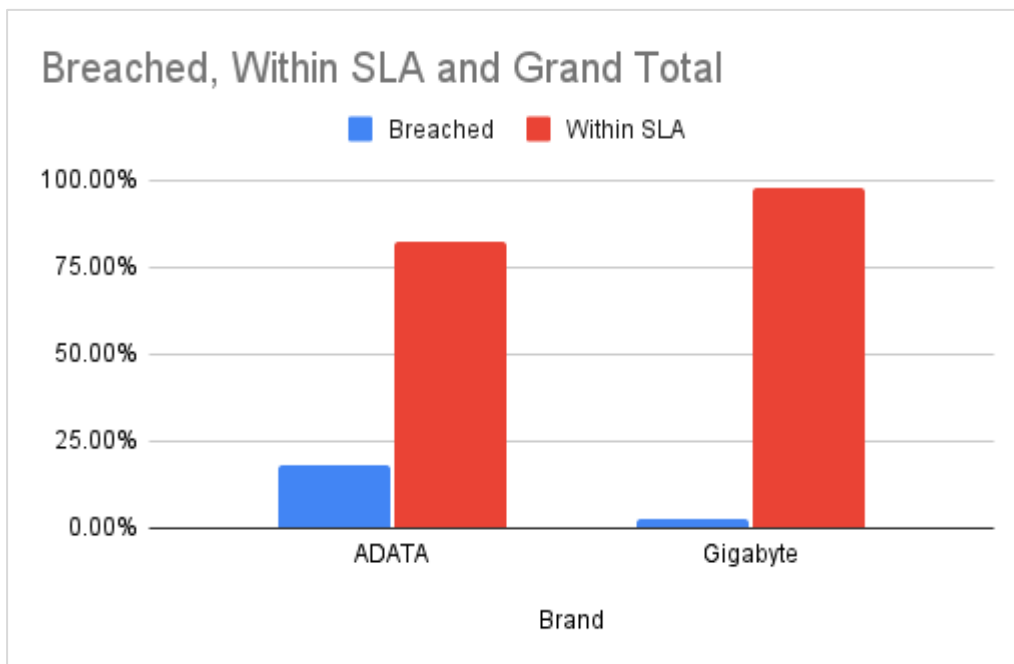
The cleaned and consolidated dataset contains the following columns. Each variable is defined below:

| Column Name          | Type        | Description                                               |
|----------------------|-------------|-----------------------------------------------------------|
| <b>RMA NO</b>        | Nominal     | Unique identifier for each service                        |
| <b>Product Name</b>  | Categorical | Product/model name                                        |
| <b>Brand</b>         | Categorical | Gigabyte or Adata                                         |
| <b>Receive Date</b>  | Date        | When the defective product was received for repair        |
| <b>Repair Date</b>   | Date        | When repair work was completed                            |
| <b>Shipping Date</b> | Date        | When the repaired product was shipped back                |
| <b>Close Date</b>    | Date        | When the RMA case was officially closed                   |
| <b>Send Date</b>     | Date        | When the product was initially sent to the service center |
| <b>Engineer ID</b>   | Nominal     | ID of service engineer assigned to the case               |
| <b>Month</b>         | Date        | Month grouping for analysis                               |
| <b>TAT Days</b>      | Numerical   | Total turnaround days (from receive to close)             |
| <b>Repair Days</b>   | Numerical   | Number of days spent in repair                            |
| <b>Ship Days</b>     | Numerical   | Number of days spent shipping                             |
| <b>SLA Status</b>    | Categorical | Compliance with SLA: 'Breached' or 'Within SLA'           |

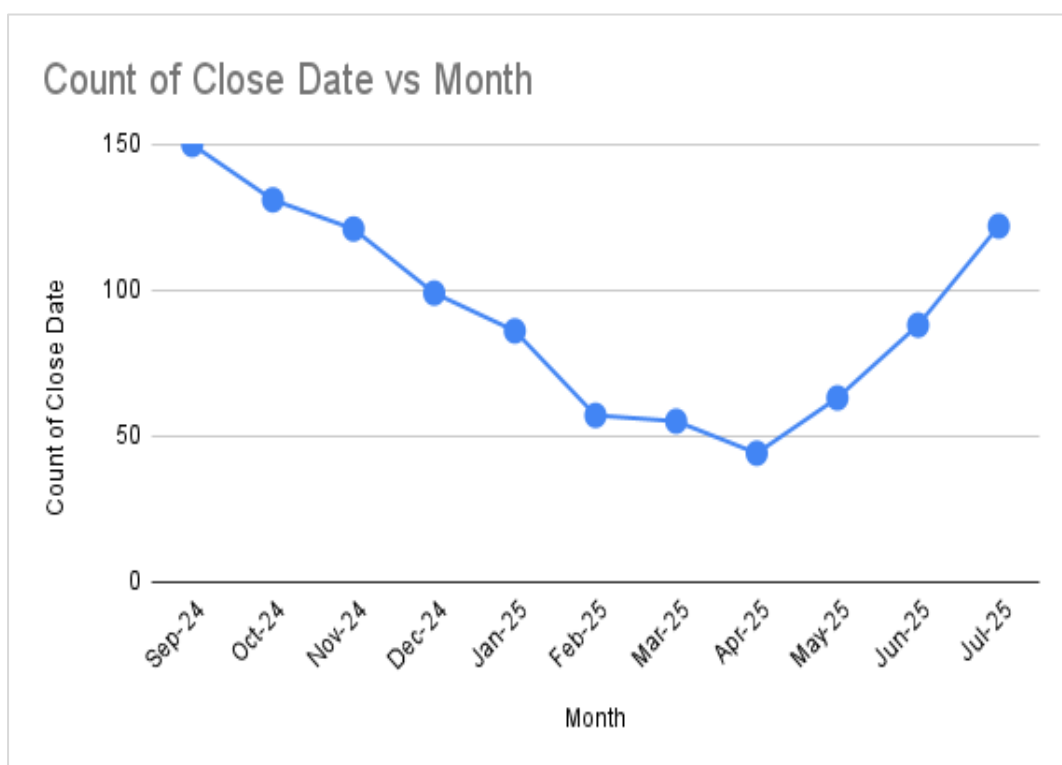
Descriptive statistics analysis follows:



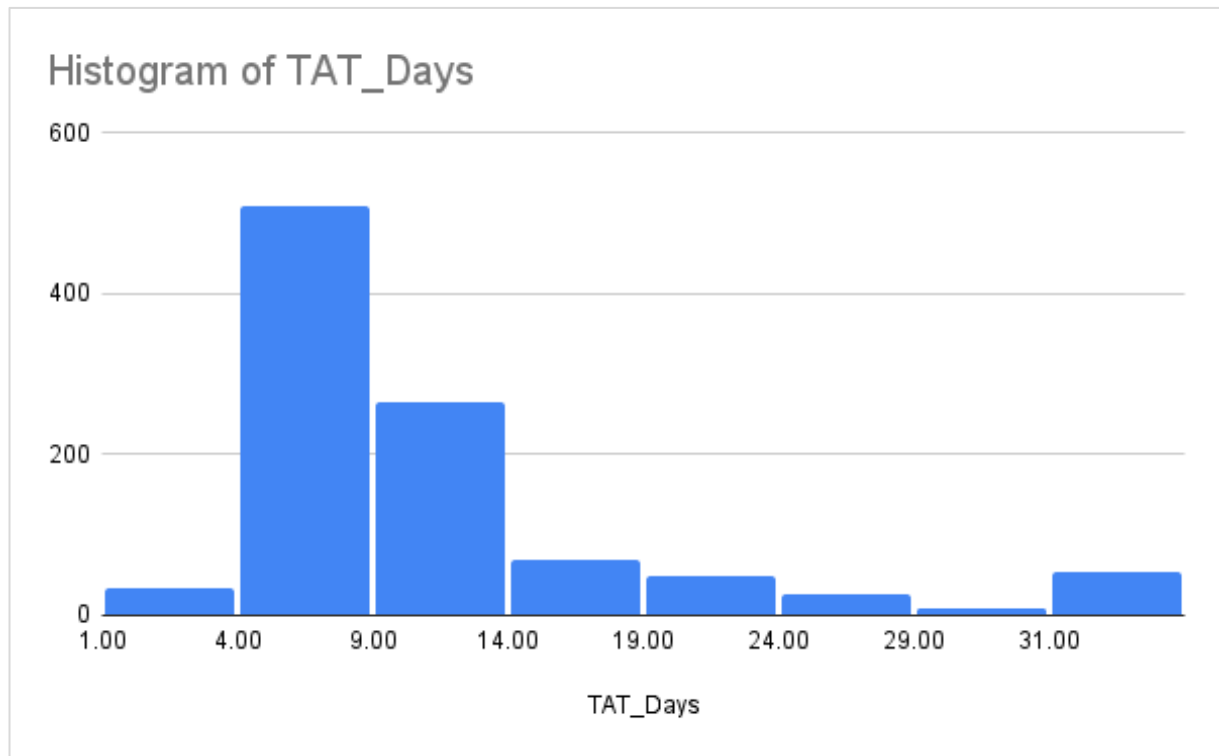
The average turnaround time (TAT) in days for warranty claims by brand (Gigabyte vs ADATA).



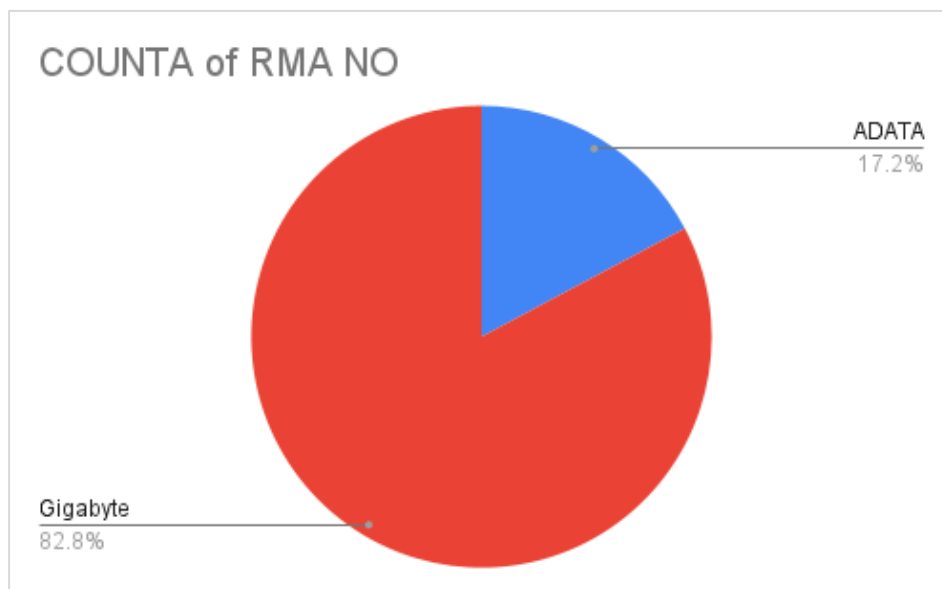
The percentage of claims breaching SLA versus those completed within SLA for each brand



Monthly trends in the count of closed warranty service cases across the observed period



The distribution of turnaround times (TAT) for all warranty claims in days



The proportion of total warranty claims handled for Gigabyte versus ADATA



## **4 Detailed Explanation of Analysis Process/Method :**

Data for the warranty service analysis was collected and compiled from Excel and Google Sheets records over an 11-month period. Each month's data was merged, standardized, and quality-checked to create a single master dataset. Initial steps involved renaming columns for consistency, removing duplicate entries, reconciling mismatched workflow dates, and excluding records with incomplete essential information. Additional key dates, such as send dates, were cross-referenced and integrated from supplementary files to ensure a complete timeline for every service case.

Once the data integrity was established, new columns were generated to support service process evaluation. These included: turnaround time (TAT Days), calculated as the difference between receive and close dates; repair days and shipping days, providing more granular visibility of each case; and SLA status, coded based on whether each task met the business's 30-day resolution benchmark. The dataset was further enriched with brand, engineer, and monthly grouping attributes.

With the enhanced dataset, descriptive statistics were computed to summarize processing efficiency and compliance. Key metrics included average turnaround time (TAT) by brand, total counts and percentages of cases breaching the 30-day SLA, and monthly volume of closed warranty claims. These statistics enabled a clear understanding of service consistency and resource utilization patterns over time and across brands.

Data visualization played a pivotal role in interpreting results, using several chart types matched to specific analytical questions. Bar charts illustrated average TAT differences between Gigabyte and ADATA, highlighting brands with longer service durations. Column charts emphasized breach counts and percentages, allowing straightforward reliability comparisons. Distribution patterns of TAT were explored through histogram. Line charts traced monthly trends in service volume to detect seasonal fluctuations, particularly the notable summer dip. Finally, pie charts conveyed brand share in total claims, clarifying dependence on key partners.

Together, these analyses and visualizations formed a comprehensive understanding of Prime Infotech's operational challenges, and helped identify opportunities to reduce SLA breaches, balance workloads throughout the year, and improve technician productivity. This methodical approach ensured the midterm report provides actionable, data-driven insights required to optimize warranty service delivery and support strategic decision-making.

While the detailed analysis centers on Gigabyte and ADATA, which maintain complete and date wise claim data, Prime Infotech does handle other brands such as D-Link, Digisol, Intel,

and Seagate. However, for these brands, only the total number of products serviced each month is recorded without receive, repair, ship, or close dates. As these brands have not set up systems to record workflow level service dates, metrics such as turnaround time and SLA breach rates cannot be calculated or included. Therefore, all charts, statistics, and findings in this report reflect only those cases and brands with sufficient workflow detail, ensuring accurate benchmarking for SLA compliance and process reliability.

## 5 Results and Findings

- Turnaround Time Distribution:

The histogram of TAT Days shows that a majority of warranty claims are resolved within 10–20 days, reflecting an efficient standard for most cases. However, there are a small number of claims with much longer turnaround times, indicating outliers or complex cases that require investigation. This pattern suggests overall process reliability with some exceptions driving SLA breaches.

- Brand Comparison of TAT:

Bar chart analysis reveals that Gigabyte typically has a lower average turnaround time compared to ADATA. This implies that warranty claims for Gigabyte products are processed and resolved more quickly, highlighting brand-based operational differences. ADATA claims tend to take longer, potentially leading to a higher breach rate.

- SLA Breach Rates:

Column charts of SLA status show that SLA breaches (turnaround time over 30 days) are more frequent for ADATA than Gigabyte. This points to ADATA requiring a focused process review. Overall, the majority of claims are closed within SLA, but a non-trivial portion consistently exceeds the compliance threshold.

- Monthly Service Volume Trends:

The line chart of monthly closed claims demonstrates periodic fluctuations in service workload, with a notable dip in summer months. This seasonal decline results in underutilized technician capacity and reduced revenue for the company during this period.

- Brand Share in Total Claims:

A pie chart of brand distribution illustrates that Prime Infotech's operations are heavily dependent on Gigabyte, which accounts for the larger percentage of warranty claims compared to ADATA. This exposes a business risk, as market or policy changes with Gigabyte would significantly impact service volume and revenue.